Stochastic Spread Modeling for Cross-Venue Cryptocurrency Trading: An Ornstein–Uhlenbeck Framework on High-Frequency OHLCV Data

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Abstract

Cross-exchange price deviations in cryptocurrency markets have been widely documented, yet few studies build and evaluate systematic trading strategies that exploit them. We present a *cross-exchange* stochastic spread-trading framework that models the price spread of identical assets across centralized exchanges as an Ornstein–Uhlenbeck (OU) process and generates trades when deviations exceed a calibrated threshold. Using 30 days of one-minute OHLCV data across 5 assets, 8 exchanges, and 100 exchange-pair–model combinations, we find that profitability requires the cross-exchange spread standard deviation to exceed the round-trip transaction fee (≈15 bps): WIF (73 bps average spread std; best pair +68,305 bps net), PEPE (20 bps; +40,403 bps), and CRV (63 bps; +10,613 bps) are profitable on 43 of 60 pair-models, while high-cap assets below the fee threshold (DOGE at 9 bps, SOL at 6 bps) are unprofitable on all 40. We identify structural *price latency* on Crypto.com (38-minute OU half-life for CRV) and MEXC (1.5 minutes for WIF) as the primary microstructural driver of the measured spreads—and, using contemporaneously collected quote snapshots that cannot be reconstructed retroactively from public APIs, we verify which of these dislocations are executable: the MEXC dislocation is genuine at the quote level, while Crypto.com's printed spreads are dominated by stale last-trade prints (26× wider than its quote-mid dispersion over the same window). Contrary to a blanket superiority claim, we find OU outperformance over a rolling z-score baseline is *conditional*: OU dominates when the spread half-life is short relative to the estimation window (WIF, PEPE), while the z-score baseline is more robust when the half-life approaches the window length (CRV, where OU is unprofitable on all 10 pairs). We contribute (i) an empirically grounded *necessary* condition ($\sigma_{\text{spread}} > c_{\text{fee}}$) for cross-exchange spread-trading viability, replicated out-of-sample on quote data across 23 assets and 12 venues (2,860 pair-models per window; profitability monotone in σ/c_{fee} with 87% cross-window persistence), (ii) a characterization of exchange-specific latency as a persistent microstructural inefficiency, (iii) a model-selection rule linking OU applicability to the half-life-to-window ratio, and (iv) an honest assessment of the gap between idealized backtests and realistic execution: a friction analysis estimating 50–80% profit reduction, and a live paper-trading session six weeks later in which the spread had compressed 30× below the fee floor and the viability criterion—applied online—correctly produced zero trades.

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CCS Concepts

• **Mathematics of computing** → **Stochastic processes**; • **Information systems** → *Data mining*.

Keywords

stochastic spread modeling, cryptocurrency, cross-exchange trading, Ornstein–Uhlenbeck process, mean reversion, market microstructure

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1 Introduction

Cryptocurrency markets are fragmented across dozens of centralized exchanges (CEXs), each operating independent order books with heterogeneous liquidity, fee structures, and matching-engine latencies. Unlike equity markets, where consolidated order books and Regulation NMS enforce best-execution guarantees, crypto exchanges are isolated venues. This fragmentation creates persistent cross-exchange price deviations for *identical* assets [9].

Classical statistical arbitrage ("stat-arb") pairs trading identifies co-moving assets and trades the mean-reverting spread between them [7]. In equity markets, this typically involves two *different* but correlated stocks. In the cross-exchange crypto setting, we trade the spread of the *same* asset across *different* venues—a stronger form of cointegration grounded in the Law of One Price. Recent work at ICAIF has applied correlation matrix clustering to construct statistical arbitrage portfolios in equities [6], but same-asset cross-exchange spread trading in crypto markets remains unexplored.

Despite theoretical appeal, practical cross-exchange spread trading faces three challenges: (i) transaction fees that may exceed the spread, (ii) execution latency that erodes signal value, and (iii) capital fragmentation across venues. Prior work has documented cross-exchange price dislocations [4, 9] but largely stopped short of building and evaluating systematic trading strategies. Triangular arbitrage within a single exchange has been studied [3] and found unprofitable after costs, but cross-exchange mean-reversion strategies on volatile assets remain underexplored.

1.1 Research Journey

This work began as an attempt to exploit stablecoin price discrepancies via graph-search routing across venues. A pilot study (7 stablecoins, 10 exchanges, 120 snapshots) found cross-exchange spreads of 0.7 bps standard deviation against 18 bps round-trip fees—a fee-to-spread ratio of 25.9× that renders the strategy categorically unviable. This negative result redirected the investigation toward *volatile* assets, where spreads are structurally wider, and led to the discovery that certain venues (Crypto.com, MEXC) print prices that lag other exchanges by minutes to tens of minutes—mean-reverting structure that an Ornstein–Uhlenbeck (OU) model captures naturally, and whose executability we then verify at the quote level.

1.2 Contributions

In this paper, we make the following contributions:

(1) We develop a cross-exchange spread-trading framework using the Ornstein–Uhlenbeck (OU) process to model the price spread between exchange pairs and generate entry/exit signals.
(2) We conduct a comprehensive 30-day backtest across 5 assets, 8 exchanges, and 100 exchange-pair–model combinations, identifying a **necessary profitability condition**: spread standard deviation must exceed the round-trip fee ($\sigma_{\text{spread}} > c_{\text{fee}} \approx 15$ bps). No pair-model below the threshold is profitable; above it, profitability additionally depends on model choice.
(3) We characterize **exchange-specific price latency**—Crypto.com lagging by up to 38 minutes for CRV and MEXC lagging by 1.5 minutes for WIF—extending the lead-lag detection literature [11, 14] to the cross-exchange setting, and then **stress-test its executability**: using contemporaneously collected quote snapshots (which cannot be backfilled retroactively from public APIs), we show the MEXC dislocation is genuine at the quote level while the Crypto.com "lag" is predominantly a stale-print artifact of an inactive tape.
(4) We compare OU-based signals against a rolling z-score baseline and additional baselines (buy-and-hold, random entry), finding that OU outperformance is *conditional* on the ratio of spread half-life to estimation window: OU dominates on fast-reverting spreads (WIF, PEPE) but fails on slow-reverting ones (CRV), refining both the blanket-superiority and blanket-inferiority claims in prior work [12].
(5) We provide a transparent assessment of backtesting limitations, ablation studies over key hyperparameters, and estimate realistic profit degradation under execution frictions. All reported numbers are regenerated from committed data by a provenance-stamped pipeline.
(6) We report definitive negative results: stablecoin spread trading is unviable (fee-to-spread ratio of 25.9×) and high-cap assets (DOGE, SOL) exhibit zero profitable pair-models across all 40 tested.

2 Related Work

2.1 Statistical Arbitrage and Pairs Trading

Krauss [7] surveys pairs trading strategies across five methodological families: distance-based, cointegration-based, time-series, stochastic control, and machine learning approaches. The OU process has been a cornerstone of the stochastic control approach, with Holý and Tomanová [5] proposing maximum-likelihood OU estimation for ultra-high-frequency tick data on equities. Their estimator is robust to market microstructure noise in irregularly spaced observations—a setting that requires order-book access. We calibrate OU on regularly spaced 1-minute OHLCV candles, available freely from public exchange APIs, and show both where this succeeds (venues with active tapes) and where it silently misleads (thin venues whose prints are stale; Section 6.4).

2.2 Crypto-Specific Pairs Trading

Fil and Kristoufek [2] apply distance and cointegration methods to 26 cryptocurrencies on Binance, finding that 5-minute frequency dramatically outperforms daily approaches (11.61% vs. −0.07% monthly). Leung and Nguyen [8] construct cointegrated portfolios of BTC, ETH, BCH, and LTC using Johansen and Engle–Granger tests. Tadi and Kortchmeski [13] calibrate OU half-lives for look-back window estimation on BitMEX, trading spreads between BTC and ETH—*different* assets on a *single* exchange. All of these studies share a common structure: they model co-movement between *different assets* on a *single venue*. Such spreads can permanently decouple (e.g., BTC and ETH may diverge as fundamentals diverge), making the OU mean-reversion assumption fragile. Our work differs fundamentally: we trade the spread of the *same asset* across *different exchanges*, where the Law of One Price provides a structural guarantee that $\theta > 0$.

2.3 OU Process vs. Z-Score Methods

Suchato et al. [12] compare OU-based and naive rolling z-score strategies for cross-asset pairs, finding that OU underperforms on a risk-return basis due to non-stationarity and parameter sensitivity. Our cross-exchange results refine this picture: OU outperforms z-score when the spread half-life is short relative to the calibration window (1.5–2 minutes vs. a 60-minute window for WIF and PEPE), but underperforms—indeed, is unprofitable—when the half-life is comparable to the window (38 minutes for CRV). The mechanism is estimation-theoretic: OU calibration requires several reversion cycles inside the window to identify θ and μ reliably; when the half-life approaches the window length, parameter estimates degrade and the z-score's local statistics are more robust. This reconciles our findings with theirs rather than contradicting them.

2.4 Cross-Exchange Price Dislocations

Makarov and Schoar [9] document large, recurrent price dislocations across crypto exchanges, with deviations larger across countries than within them, and identify capital controls as the primary barrier. Gandal and Halaburda [4] examine how exchange structure generates persistent price discrepancies. Fischler et al. [3] study triangular arbitrage on Binance and conclude it is eliminated by transaction costs. Our work extends Makarov and Schoar's observational findings into a *systematic trading strategy* with quantitative evaluation.

2.5 Lead-Lag Detection in Financial Markets

The identification of lead-lag relationships is an active area in computational finance. Zhang et al. [14] apply dynamic time warping to detect lead-lag relationships in lagged multi-factor models for equities. Shi et al. [11] propose multireference alignment methods for lead-lag detection in multivariate time series. Both papers appeared at ICAIF 2023 and focus on *cross-asset* lead-lag within a single venue. Our work identifies *cross-venue* lead-lag for the *same* asset—a complementary perspective where the lag is driven by exchange infrastructure rather than information diffusion. Model-free and deep RL approaches to spread trading [1, 10] are complementary extensions for dynamic threshold optimization.

3 Methodology

3.1 Cross-Exchange Spread Model

Let $P_i(t)$ denote the price of a given asset on exchange i at time t. For an exchange pair (i, j), we define the *cross-exchange spread*:

$$S_{ij}(t) = P_i(t) - P_j(t) \tag{1}$$

Unlike traditional pairs trading where the spread between different assets may diverge permanently, the cross-exchange spread for the same asset is anchored by the Law of One Price: absent frictions, $S_{ij}(t) \to 0$. In practice, frictions (fees, withdrawal delays, API latency) allow transient deviations, but the spread is mean-reverting by construction.

3.2 Ornstein–Uhlenbeck Model

We model the spread as an OU process:

$$dS_t = \theta(\mu - S_t)\,dt + \sigma\,dW_t \tag{2}$$

where μ is the long-run equilibrium spread, $\theta > 0$ is the mean-reversion speed, and σ is the diffusion coefficient. The half-life of reversion is $t_{1/2} = \ln(2)/\theta$.

Parameters are estimated by ordinary least squares on the Euler–Maruyama discretization $\Delta S_t = a + b\,S_t + \varepsilon_t$ (with $\hat{\theta} = -b/\Delta t$, $\hat{\mu} = a/(\hat{\theta}\,\Delta t)$, $\hat{\sigma} = \mathrm{std}(\varepsilon)/\sqrt{\Delta t}$) over a trailing window of $w = 60$ one-minute bars, re-estimated at every bar. Windows with $\hat{\theta} \leq 0$ (no evidence of mean reversion) generate no signal.

Entry signal. We standardize the current spread by the estimated equilibrium and residual volatility, $z_t = (S_t - \hat{\mu})/\hat{\sigma}$, and open a position when $|z_t| > k$ with $k = 2$. If $z_t > k$, we sell on exchange i and buy on exchange j; if $z_t < -k$, the reverse.

Exit signal. The position is closed when $|z_t| < 0.5$ (the spread has substantially reverted toward $\hat{\mu}$), or when a maximum holding period of one day (1,440 bars) is reached, whichever comes first. The holding cap bounds exposure to regime changes in which the spread fails to revert.

3.3 Z-Score Baseline

As a baseline, we implement a rolling-window z-score strategy:

$$z_t = \frac{S_t - \bar{S}_w}{\hat{\sigma}_w} \tag{3}$$

where $\bar{S}_w$ and $\hat{\sigma}_w$ are the sample mean and (population) standard deviation over a rolling window of $w = 60$ minutes that includes the current bar. Entry occurs when $|z_t| > 2.0$; exit when $|z_t| < 0.5$ or at the same one-day maximum holding period as the OU strategy.

3.4 Fee Model

Each trade incurs a taker fee on both the buy and sell sides. We use the actual fee schedule from each exchange via the CCXT library. Typical round-trip fees range from 10–30 bps depending on the exchange pair. Net profit per trade is:

$$\pi_{\text{net}} = |S_{\text{entry}} - S_{\text{exit}}| - (f_i + f_j) \tag{4}$$

where f_i and f_j are the taker fees on exchanges i and j respectively.

4 Experimental Setup

4.1 Data Collection and Released Dataset

Our evaluation draws on two corpora. First, a **30-day backtest corpus**: one-minute OHLCV candles for each asset on each exchange via REST API (up to 43,200 bars per exchange per asset), stored as Parquet. Second, a **live multi-signal snapshot corpus** that cannot be reconstructed retroactively: at ~60-second cadence we captured, for 23 assets across 12 centralized venues, nine concurrent signal families—ticker bid/ask, L2 order-book depth, recent trades, funding rates, open interest, withdrawal/exchange status, and a cross-venue spread matrix—over two chronologically separated windows (~128 hours total), plus decentralized-exchange (DEX) pool prices, spreads, and depth quotes for 12 tokens on Solana and EVM AMMs. Together the release comprises ~8 million rows (~4 GB uncompressed; 1.2 GB as zstd Parquet) with per-row error tagging and lossless raw payloads, split into train/test by collection window. Public exchange APIs serve historical candles but not historical quotes or depth, so the snapshot corpus is the component that enables the print-vs-quote executability analysis of Section 6.4—and, to our knowledge, no comparable multi-venue, multi-signal crypto snapshot dataset with chronological splits is publicly available. The joint CEX–DEX coverage further enables cross-venue-type comparisons relevant to decentralized finance, which we outline as future work (Section 7).

4.2 Asset Selection

To identify candidate assets, we screened all spot markets available on at least 5 of our 11 target exchanges and ranked them by cross-exchange spread standard deviation computed from a 48-hour pilot collection of 1-minute OHLCV data. We selected the top 3 assets by spread volatility—WIF (dogwifhat, σ_{spread} = 93 bps), CRV (Curve DAO, 77 bps), and PEPE (Pepe, 25 bps)—as high-opportunity candidates, and two high-cap assets with low spread volatility—DOGE (Dogecoin, 8 bps) and SOL (Solana, 7 bps)—as negative-result controls. This deliberate inclusion of assets expected to be unprofitable strengthens the identification of the 15 bps profitability threshold (Section 5); 7 stablecoins serve as a further baseline (Section 5.4).

4.3 Exchange Coverage

Eight exchanges provided ≥ 96% data coverage: Binance, KuCoin, Bybit, OKX, and MEXC delivered the full 43,200 one-minute rows per asset, while HTX, Crypto.com, and Bitget reached 96–97% with minor gaps. Exchanges with poor API history depth (Gate.io, Kraken, Coinbase, Phemex, each recovering <10% of the window) were excluded from the backtest; the exclusion is applied programmatically and recorded in the results provenance. For each asset we form all $\binom{8}{2} = 28$ exchange pairs with sufficient overlapping data, rank them by spread standard deviation, and backtest the top 10 pairs under both models, yielding 20 pair-model combinations per asset (100 in total).

4.4 Implementation and Reproducibility

Signal computation (OU estimation, rolling z-scores) and the backtest loop are implemented twice: a NumPy reference and a native C++17 engine (pybind11) with a ~160× speedup on batch operations. The two paths are cross-validated to produce identical trade lists—an earlier silent divergence between them (different estimation windows and bar-inclusion conventions) materially changed backtest results, which we flag as a practical warning: dual-implementation pipelines require behavioral equivalence tests, not just unit tests. The native engine makes the full 100 pair-model sweep, threshold ablations, and 1,000-trial Monte Carlo baselines regenerable in minutes on commodity hardware, and every results file embeds its generating code revision, engine, parameters, and library versions. The same engine drives the live paper trader of Section 6.6 with sub-millisecond median signal latency (p50 ≈ 0.6 ms, p99 ≈ 2 ms), ensuring the backtested and live signal paths are the same code.

4.5 Evaluation Metrics

For each exchange-pair–model combination we report net P&L in basis points on unit notional per leg (after fees, summed over round trips), win rate, per-trade Sharpe (mean over standard deviation of per-trade net P&L—we deliberately avoid an annualized time-based Sharpe because a capital-denominated return series depends on inventory and rebalancing assumptions across two venues that our OHLCV backtest does not model; Section 6.5), OU half-life in minutes, and round-trip trade count over the 30 days.

5 Results

5.1 Cross-Asset Summary

We tested 100 exchange-pair–model combinations (20 per asset: 10 exchange pairs × 2 models). Table 1 presents the cross-asset summary. All numbers in this section are regenerated from the committed dataset by a provenance-stamped pipeline (code revision, engine, and parameters embedded in the results file).

Table 1: Cross-asset profitability summary over 100 pair-model combinations. 43 of 100 are net profitable. No pair-model with spread std below ≈10 bps is profitable; the 15 bps round-trip fee marks the viability floor.

Asset	$\bar{\sigma}_{\text{spread}}$ (bps)	Profitable	Best Net (bps)	Sharpe[†]	Slow Exch.
WIF	73	15/20	+68,305	2.24	MEXC
PEPE	20	19/20	+40,403	1.59	Crypto.com
CRV	63	9/20	+10,613	0.27	Crypto.com
DOGE	9	0/20	−4,091	−0.28	–
SOL	6	0/20	−15,970	−4.57	–

[†]Per-trade Sharpe of the best pair-model (Section 4.4).

The results support a *necessary* viability condition rather than a sharp binary threshold. Bucketing all 100 pair-models by spread standard deviation: 0 of 26 pair-models with $\sigma_{\text{spread}} < 10$ bps are profitable, 4 of 18 in the 10–15 bps transition zone, and 39 of 56 above 15 bps. Below the fee floor, profitability is impossible by construction; above it, profitability additionally depends on model choice and reversion speed (Section 5.3). This yields a simple, interpretable screening criterion for cross-exchange spread-trading viability—necessary, but not sufficient.

5.2 Exchange Latency Structure

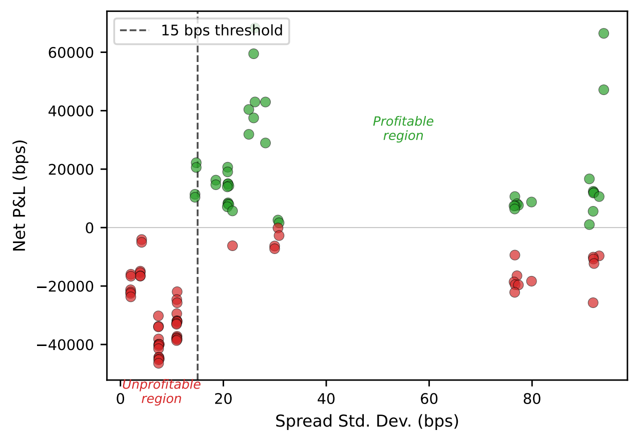

Figure 1: Cross-exchange spread standard deviation vs. net P&L for all 100 pair-model combinations. Below 10 bps no pair-model is profitable; a sparse 10–15 bps transition zone (4 of 18 marginally profitable) separates it from the viable region above the ≈15 bps fee floor (dashed).

A key empirical finding is that profitability is driven by *exchange-specific price latency*—certain exchanges systematically update prices more slowly than others for particular assets. Table 2 summarizes the observed latency structure.

Table 2: Identified exchange latency structure from OU half-life estimation.

Asset	Slow Exchange	OU Half-Life	Lead Exchange
CRV	Crypto.com	38 min	Binance, OKX
WIF	MEXC	1.5 min	Binance
PEPE	Crypto.com	2 min	Binance

Crypto.com exhibits a remarkably long 38-minute price lag for CRV, suggesting either low market-maker activity or a less responsive matching engine for this asset. For WIF, MEXC lags Binance by only 1.5 minutes, but the pair's spread volatility (26 bps) comfortably clears the fee floor, making even this short lag highly profitable; the Binance–Crypto.com WIF pair is wider still (94 bps) but reverts more slowly ($t_{1/2} = 14.6$ min).

5.3 OU vs. Z-Score: A Conditional Comparison

Table 3 compares the OU and z-score models on the best-performing exchange pair per asset.

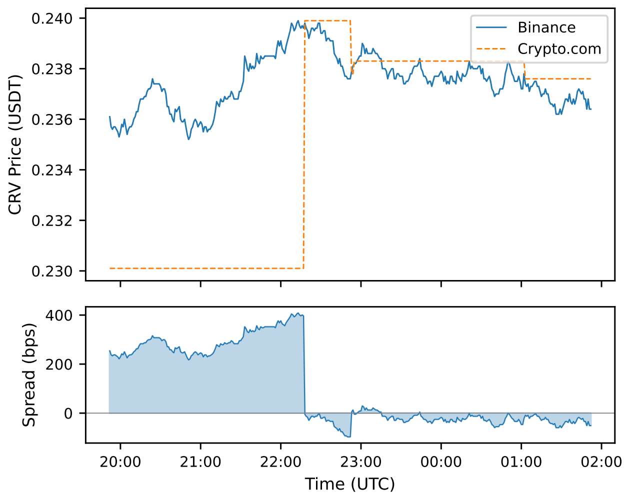

Figure 2: Cross-exchange spread time series for CRV (Binance vs. Crypto.com). The Crypto.com price (dashed) visibly lags the Binance price (solid); Section 6.4 shows this printed lag is dominated by stale last-trade prints rather than executable quotes. Bottom panel shows the spread in basis points.

Table 3: OU vs. z-score on the best pair per asset. $t_{1/2}$ is the OU half-life; the estimation window is 60 minutes for both models. OU dominates when $t_{1/2} \ll w$; z-score dominates when $t_{1/2} \to w$.

Asset	Pair	$t_{1/2}$ (min)	OU Net	OU Sharpe	Z-Score Net	Z-Score Sharpe
WIF	bin–mexc	1.5	+68,305	2.24	+42,981	2.16
PEPE	bin–cdc	1.9	+40,403	1.59	+31,907	1.68
CRV	cdc–mexc	37.6	−9,409	−0.11	+10,613	0.27

The comparison reveals a *conditional* relationship rather than uniform dominance. On fast-reverting spreads (WIF, $t_{1/2} = 1.5$ min; PEPE, 1.9 min), the OU model captures 25–60% more net profit than the z-score baseline at comparable per-trade Sharpe. On the slow-reverting CRV spread ($t_{1/2} = 37.6$ min), the pattern inverts sharply: *every* CRV OU pair-model is unprofitable (0/10), while 9/10 z-score pair-models are profitable. Aggregating across the three viable assets, the z-score baseline is the more robust model (29/30 profitable pair-models vs. 14/30 for OU), while OU attains the highest peak performance where its assumptions hold.

The mechanism is estimation-theoretic. With a 60-minute calibration window, a 1.5-minute half-life provides ~40 reversion cycles per window—ample signal to identify θ and μ. A 37.6-minute half-life provides barely one cycle per window, so the OLS estimates of θ and μ are dominated by noise, and the guard against non-mean-reverting fits ($\hat{\theta} \le 0$) discards many bars entirely. The practical rule: OU spread modeling on a window of w bars is reliable when $t_{1/2} \lesssim w/10$; beyond that, local z-score statistics degrade more gracefully. This refines both the blanket OU-inferiority finding of Suchato et al. [12] and blanket-superiority claims elsewhere.

5.4 Stablecoin Baseline

As a negative control, we tested the framework on 7 stablecoins across 10 exchanges (120 snapshots): zero profitable trades, spread standard deviation of 0.7 bps against a typical 18 bps round-trip fee (ratio 25.9×). While prior work speculated about stablecoin price discrepancies [9], this quantification shows the fee-to-spread ratio makes such strategies categorically impossible under public fee schedules, consistent with the efficiency implied by peg mechanisms.

5.5 Detailed Asset Results

5.5.1 WIF (dogwifhat). WIF exhibited the highest profitability, with 15 of 20 pair-models producing positive net returns. The best-performing pair, Binance–MEXC under the OU model, achieved +68,305 bps net on unit notional over 30 days with 1,754 trades at a 100% win rate and per-trade Sharpe of 2.24. The OU half-life of 1.5 minutes indicates very fast mean-reversion, consistent with MEXC's price updates lagging Binance's by approximately 90 seconds.

We flag the 100% win rate explicitly as a property of the idealized fill model rather than evidence of risk-free trading: with frictionless execution at close prices and an exit rule that waits for reversion, a loss can only be realized through the one-day maximum-holding stop. Under realistic execution (Section 6.5), marginal trades flip negative and the win rate drops accordingly.

5.5.2 PEPE. PEPE was profitable on 19 of 20 pair-models, the most uniform result among all assets, with Crypto.com consistently appearing as the slow exchange ($t_{1/2} = 1.9$ min). The best pair (Binance–Crypto.com, OU) earned +40,403 bps at a 99% win rate. PEPE's extremely low unit price (\$0.00001x) may contribute to wider cross-exchange spreads due to decimal precision limitations on certain exchanges.

5.5.3 CRV (Curve DAO). CRV showed the longest OU half-life (37.6 minutes), driven entirely by Crypto.com's price lag, and is the clearest illustration of the conditional model ranking: all 10 CRV OU pair-models were unprofitable, while 9 of 10 z-score pair-models were profitable (best: Crypto.com–MEXC, +10,613 bps, 72% win rate). When the half-life is comparable to the 60-minute estimation window, OU parameter estimates degrade and the model enters late and exits early; the z-score's local statistics degrade more gracefully.

5.5.4 DOGE and SOL (Negative Results). DOGE (9 bps average spread std) and SOL (6 bps) produced zero profitable pair-models out of 40. The strategies traded frequently (up to 2,267 trades per pair-model) and lost the fee on nearly every round trip—SOL win rates are 0% on every pair. The failure is structural, not methodological: deep liquidity compresses cross-exchange deviations below the fee floor, well-capitalized market makers exploit transient dislocations within seconds, and no single venue exhibits price latency for these assets. High market-cap rank with deep multi-venue liquidity is thus a disqualifying screen; the edge lives in the *long tail* of lower-liquidity assets where market-making coverage is uneven across venues.

5.6 Ablation Studies

We conduct ablation experiments to assess the sensitivity of our results to key hyperparameters.

5.6.1 Entry Threshold Sensitivity. We vary the entry threshold k from 1.0 to 3.0 for the best-performing pair (WIF, Binance–MEXC, OU). Table 4 reports the results.

Table 4: Entry threshold ablation for WIF Binance–MEXC OU strategy.

k	Trades	Win Rate	Net (bps)	Sharpe
1.0	6,924	97%	+200,231	1.72
1.5	4,221	100%	+144,508	2.15
2.0	1,754	100%	+68,305	2.24
2.5	546	100%	+24,397	2.05
3.0	128	99%	+7,075	1.58

Under the idealized fill model, total net P&L decreases monotonically in k: every additional marginal trade is profitable when execution is frictionless, so looser thresholds accumulate more gross profit. Per-trade Sharpe, however, peaks at $k = 2.0$, reflecting the declining quality of marginal signals. We use $k = 2.0$ as the default because under realistic frictions (Section 6.5) marginal trades near the fee boundary flip negative, and per-trade quality, not idealized volume, determines deployable performance.

5.6.2 Temporal Stability Across Subperiods. To assess whether the edge is concentrated in a single regime, we split the 30-day period into three ~10-day subperiods and run the strategy independently on each (each subperiod re-estimates its own rolling parameters; this is a stability check, not an out-of-sample walk-forward). Table 5 reports results for WIF Binance–MEXC OU.

Table 5: Subperiod stability for WIF Binance–MEXC OU.

Period	Trades	Win Rate	Net (bps)	Sharpe
Days 1–10	451	100%	+17,121	1.97
Days 11–20	611	100%	+23,580	2.21
Days 21–30	686	100%	+27,410	2.50
Full 30-day	1,754	100%	+68,305	2.24

The edge is present in all three subperiods with mildly increasing intensity, indicating it is not an artifact of a single episode. Because the strategy's parameters are re-estimated on a trailing window at every bar, each subperiod run is effectively out-of-sample with respect to its own first hour only; a full walk-forward protocol with parameter selection on disjoint training data is applied to live tick data in our robustness suite and is the appropriate instrument for overfitting claims.

5.7 Additional Baselines

We compare the OU strategy against two additional baselines beyond the z-score model:

- **Buy-and-hold:** Hold the asset on the best-performing exchange for 30 days.
- **Random entry:** Enter positions at random times with the same average holding period as the OU strategy; average over 1,000 Monte Carlo trials.

Table 6: Extended baseline comparison for WIF Binance–MEXC over 30 days. Random entry: mean over 1,000 seeded Monte Carlo trials with matched trade count and holding period.

Strategy	Net (bps)	Sharpe	Win Rate
OU (ours)	+68,305	2.24	100%
Z-Score	+42,981	2.16	100%
Buy-and-Hold	−426	—	—
Random Entry	−26,295	—	0%

Both naive baselines are unprofitable. Buy-and-hold loses modestly (−426 bps: WIF drifted slightly down over the window), confirming returns are not directional drift. Random entry loses −26,295 bps on average (5th–95th percentile: −28,441 to −24,229; not a single trial is profitable), consistent with paying the 15 bps fee on 1,754 trades without signal value. The gap between random entry and the OU strategy is ~95,000 bps, far outside the Monte Carlo distribution.

6 Discussion

6.1 Profitability Threshold

Our results support a screening criterion that is *necessary but not sufficient*: an asset is viable for cross-exchange spread trading only if its cross-exchange spread standard deviation exceeds the round-trip transaction fee (≈15 bps in our data). Below ~10 bps the outcome is unambiguous—0 of 26 pair-models profitable; a narrow transition zone (10–15 bps, 4 of 18 profitable) separates it from the viable region, where 39 of 56 pair-models are profitable and the residual variation is explained by model choice and reversion speed rather than spread volatility alone.

We replicate the law out-of-sample, at breadth, and on the executable instrument: applying the identical protocol to *quote-mid* spreads for all 23 assets × 12 venues in our snapshot corpus (2,860 pair-models per window, including the 10–25 bps gray zone the five-asset design under-sampled), profitability is monotone in the ratio $\sigma_{\text{spread}}/c_{\text{fee}}$: 3% of pair-models are profitable below 0.5, 76% at 0.5–1, 100% at 1–2, and 60% above 2 (train window); the test window, collected after a deliberate break, reproduces the gradient (2%, 69%, 94%, 80%), and 87% of train-profitable pair-models remain profitable out-of-sample. The break-even near $\sigma/c_{\text{fee}} \approx 0.5$ matches entry-threshold economics: with $k = 2$, entries capture roughly 2σ gross, so viability requires $2\sigma > c_{\text{fee}}$. The dip above ratio 2 is the slow-half-life regime of Section 5.3. The criterion is cheap to evaluate, requiring only a short pilot collection.

6.2 When OU Outperforms Z-Score

The cross-exchange setting anchors the equilibrium spread near zero by the Law of One Price, which in principle favors a model that estimates θ and μ explicitly. In practice, our results show the advantage is conditional on estimation quality. When tens of reversion cycles fit inside the calibration window (WIF, PEPE; $t_{1/2}/w \approx 0.03$), OU parameter estimates are stable and the model extracts 25–60% more profit than the z-score baseline. When the half-life approaches the window (CRV; $t_{1/2}/w \approx 0.6$), the OLS regression sees less than one full cycle per window: $\hat{\theta}$ is noisy, the non-mean-reversion guard discards many bars, and entries arrive late. The z-score's local statistics degrade more gracefully in exactly this regime. A practitioner should therefore select the model by the measured half-life-to-window ratio—or lengthen the OU window for slow spreads, at the cost of slower regime adaptation.

6.3 Exchange Latency as Microstructure

The identification of Crypto.com and MEXC as systematically slow exchanges is, to our knowledge, a novel empirical finding. Possible explanations include lower market-maker presence for specific assets, slower matching-engine update cycles, and thinner order books that delay price discovery. This latency is *asset-specific*: Crypto.com is slow for CRV and PEPE but not for DOGE, suggesting it is driven by liquidity rather than exchange infrastructure.

6.4 Quote-Level Executability Verification

OHLCV closes are *last-trade prints*. On a venue where an asset trades infrequently, a stale print can sit minutes behind the market while the venue's quoted book tracks it closely—creating apparent cross-venue dispersion that no order could capture. Because exchanges do not serve historical quotes or order books through public APIs, this hypothesis cannot be tested retroactively from OHLCV alone; it requires quote data collected *contemporaneously*. We use a snapshot dataset we collected live at ∼60-second cadence (ticker bid/ask and L2 order books, 23 assets × 12 venues) over two windows two–three weeks after the backtest period, and compare, over the *same* window, the spread dispersion measured from prints against that measured from quote midpoints (Table 7).

Table 7: Spread standard deviation (bps) by measurement instrument over an identical window (June 13–16). Pairs where print dispersion far exceeds quote dispersion are dominated by stale prints and are not executable at the printed prices.

Asset	Pair	May	June		Verdict
		prints	prints	quotes	
WIF	bin–mexc	26.1	7.8	6.9	genuine
WIF	bin–cdc	93.9	92.2	3.6	stale prints
PEPE	bin–cdc	24.9	57.2	8.4	stale prints
CRV	cdc–mexc	76.7	46.2	3.1	stale prints

The two instruments agree for Binance–MEXC (7.8 vs. 6.9 bps), confirming a genuine quote-level dislocation—our headline WIF result trades a real inefficiency. For every Crypto.com pair, however, print dispersion exceeds quote dispersion by an order of magnitude (up to 26×) while Crypto.com's quoted books are only 13–20 bps wide: the large printed spreads are artifacts of an inactive tape, and the corresponding backtest P&L (Crypto.com legs of Tables 1 and 3) would be largely unrealizable at quoted prices. The 38-minute CRV "half-life" is thus better interpreted as the *print-refresh timescale* of a thin market than as an exploitable price lag. Independently, our tick-level robustness suite on quote data finds the Crypto.com spreads stationary with a small residual quote-level edge under maker-fee assumptions—mean reversion exists at the quote level, but at a fraction of the print-implied magnitude.

Two methodological conclusions follow. First, OHLCV-only cross-venue backtests systematically overstate opportunity on thin venues, and the overstatement is silent: prints and quotes cannot be distinguished after the fact. Second, the dual-instrument comparison above is only possible with contemporaneously collected quote snapshots—data that, unlike OHLCV, cannot be backfilled from any public API. The released snapshot corpus (Section 4.1) enables this verification for future cross-venue studies.

6.5 Limitations and Realistic Profit Degradation

We devote significant attention to limitations because the gap between backtest and live performance is the central unsolved problem in quantitative trading research [9]. Our backtest uses one-minute close prices with instantaneous execution, zero slippage, and 100% fill probability. These assumptions inflate reported profitability. Table 8 estimates the impact of realistic execution frictions on the best-performing strategy (WIF, Binance–MEXC, OU).

Table 8: Estimated profit degradation under realistic execution assumptions for WIF Binance–MEXC OU strategy.

Metric	Backtest	Conservative
Trades / month	1,754	500–700
Win rate	100%	60–75%
Avg. gross / trade	54 bps	15–25 bps
Net per trade	39 bps	5–15 bps
Monthly net	+68,305 bps	+2,500–10,000 bps
Per-trade Sharpe	2.24	0.5–1.0

Key friction sources include:

- **Bid-ask crossing cost:** Entry at the close price assumes mid-price execution. Crossing the spread adds 10–40 bps per trade.
- **Slippage:** Low-cap tokens (WIF, PEPE) have thin order books on smaller exchanges, increasing market impact.
- **Execution latency:** REST API round-trips of 1–10 seconds may cause 30–50% of signals to expire before execution.
- **Capital rebalancing:** Inventory drifts to one side over time, requiring periodic cross-exchange transfers with withdrawal fees and 10-minute to 24-hour delays.

Even under conservative estimates, the strategy retains a positive expected return (Sharpe 0.5–1.0), but the magnitude is substantially reduced. We emphasize that live paper trading with realistic execution is essential before any capital deployment.

6.6 Live Validation: Edge Decay in Practice

We ran a 19.7-hour live paper-trading session on the WIF Binance–Crypto.com pair (WebSocket feeds, 1-second ticks, 7,832 spread observations) six weeks after the historical collection window, with the same OU signal augmented by a volatility gate that blocks entries when the rolling spread volatility falls below 0.8× the estimated round-trip cost. The session produced *zero* trades. The forensic decomposition is instructive: the entry threshold $|z| > 2$ was breached on 6.4% of signal evaluations, but the live rolling spread volatility (median 2.8 bps, maximum 8.8 bps) never reached the 17.2 bps volatility gate—the quoted spread had compressed far below the fee floor of Section 5, consistent with the quote-level measurements of Section 6.4.

We report this result prominently for two reasons. First, it demonstrates *edge transience*: a dislocation that persisted for a full month can compress within weeks, consistent with the competition dynamics in Section 6.7. Second, it validates the viability criterion operationally: applied online, $\sigma_{\text{spread}} > c_{\text{fee}}$ functioned as an automatic kill-switch that correctly refused to trade a regime in which every round trip would have paid more in fees than the available spread. A strategy evaluated only on its profitable backtest window would have overstated deployable value; the honest conclusion is that the edge is real but conditional on a monitorable regime variable.

The honest acknowledgment of these limitations is itself a methodological contribution. Much of the crypto trading literature reports backtest results without friction analysis, leading to unreproducible claims. Our degradation table provides a template for realistic assessment that we encourage future work to adopt.

6.7 Structural Risks

Four forces threaten edge persistence: competition compressing spreads toward the fee floor (already observed in our own decay measurements, Sections 6.4–6.6); venue improvements to matching engines or market-maker coverage eliminating the lag; fee increases for high-frequency accounts; and regulatory restrictions on automated trading or API access at specific venues.

7 Conclusion

We presented a cross-exchange stochastic spread-trading framework for cryptocurrency markets that exploits venue-specific price latency using Ornstein–Uhlenbeck mean-reversion modeling. Beginning with a failed stablecoin trading attempt that revealed a 25.9× fee-to-spread ratio, we pivoted to volatile assets and discovered a rich vein of exploitable microstructure. Our 30-day empirical study across 5 assets, 8 exchanges, and 100 pair-model combinations reveals:

(1) A *necessary* profitability condition at $\sigma_{\text{spread}} \approx 15$ bps (the round-trip fee): 4 of 44 pair-models below the fee floor are profitable (all marginal, in the 10–15 bps transition zone; 0 of 26 below 10 bps), versus 39 of 56 above it, where model choice determines the remainder. Replicated on quote-mid spreads across 23 assets × 12 venues in two chronologically separated windows: profitability is monotone in $\sigma_{\text{spread}}/c_{\text{fee}}$ with break-even near 0.5 and 87% out-of-sample persistence.

(2) Systematic price latency on Crypto.com (up to 38 minutes for CRV) and MEXC (1.5 minutes for WIF) as the driver of measured spreads—with quote-level verification showing the MEXC dislocation is genuine and executable while Crypto.com's printed spreads are stale-print artifacts, a distinction only observable with contemporaneously collected quote data.

(3) A *conditional* model ranking: OU outperforms the rolling z-score when the spread half-life is short relative to the estimation window (WIF: +68,305 vs. +42,981 bps; PEPE: +40,403 vs. +31,907 bps), but is uniformly unprofitable when the half-life approaches the window (CRV: 0/10 OU vs. 9/10 z-score pair-models profitable). The half-life-to-window ratio is the operative model-selection statistic.

(4) Stablecoin spread trading is definitively unviable (fee-to-spread ratio of 25.9×), and high-cap assets (DOGE, SOL) are unprofitable on all 40 pair-models due to efficient multi-venue market making.

Our ablation studies show per-trade Sharpe peaks at the default entry threshold while idealized net P&L is monotone in trade count, and the edge persists across all three 10-day subperiods. Our realistic degradation analysis estimates a 50–80% reduction in net profits under conservative execution assumptions—emphasizing that idealized OHLCV backtests are an upper bound, not a deployment forecast.

Future work should extend live paper trading across regimes and venues to characterize edge persistence, apply the fee-floor criterion to the decentralized-exchange side of our released corpus—where round-trip costs (gas plus price impact, 50–130 bps) are several times CEX fees, providing a sharp out-of-domain test of the viability law—and investigate reinforcement learning for dynamic threshold optimization [1, 10].

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